Academic Paper Reviewer: A Multi-Perspective Review Skill for Computer Vision and Machine Learning Papers

Technical Report, version 2.10.1

Academic Paper Reviewer project
https://github.com/Shyandram/academic-paper-reviewer.skill

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Abstract

We describe *Academic Paper Reviewer*, a Markdown skill for Claude and Codex agents that simulates an editorial process for a manuscript. The process has a field-aware panel of five independent reviewers, a literature check that cannot find the paper under review, a report check, and an Editor-in-Chief (EIC) who weights each review by its quality and makes the decision. Scores are aggregated by a deterministic, standard-library Python script and translated into the forms of 11 venues.

We evaluate the skill in four ways.

- **Accepted papers with planted flaws.** On 30 recently accepted CVPR/ECCV/ICLR papers, 96% of cited locations, 99% of quoted phrases and all but two of 842 quoted numbers are found in the paper, and the reviewers mention all six planted fatal flaws.
- **ICLR 2026 calibration.** This follows the paperreview.ai protocol. On the balanced 86-paper set with Sonnet at high effort, the reviewers' mean rating reaches Spearman 0.47 [0.27, 0.64] with the mean human rating (two human reviewers of the same papers: 0.14) and an acceptance AUC of 0.64. On the ICLR scale the normalized mean absolute error is 0.084, with essentially no bias (NME −0.003). The 20-paper pilot gave the same ranking (0.48).
- **Weaknesses compared with the human reviews.** On the same 86 papers the panel raises 46% of the human reviewers' main issues and 74% of those two or more reviewers share; the more important the issue (consensus, a reject vote, a reviewer's lead point), the more often it is raised. The EIC's five top concerns match human issues more often than an average panel issue (34% vs 26%) but hold only a third of the consensus issues. On a 20-paper pilot, an optional Gap Scout that names concrete missing items raises 65% alone and 73% with the panel. A redesign that moved these ideas into the panel (version 2.11) raised no more issues, ranked papers worse (Spearman 0.32) and cost 3.5× as much, so it was withdrawn.
- **The same model without the skill.** On the same 86 papers, one plain review by the same model at the same effort gives 69 papers the same rating; the skill's panel mean separates them (13% of paper pairs tie against 66%; pairs ordered correctly 68% against 62%). On three papers with 15 planted defects the skill finds all 15 and the plain review 12 (it misses the numbers that must be checked across tables), and only the skill's score falls. Agreement with the human reviewers is the same (Spearman 0.47 and 0.45, AUC 0.64 and 0.63, 48% and 46% of the human main issues), but the skill states the issues more concretely: 60% of its weakness points name a place in the paper and 54% say what to fix, against 26% and 6%. The skill costs about ten times more.

A smaller model (Haiku) shows no reliable signal, and neither do two GPT-6.1 models run on the same 20 papers. We also report a negative result: NMAE alone does not measure skill on this pool, because a constant predictor scores 0.089. The main remaining weakness is compression: 78% of papers receive the same final rating.

1 Introduction

Large language models can now write fluent paper reviews. Fluency, however, is not the goal of a review. Authors need the decisive weaknesses found, grounded in the manuscript, and ranked by how much they matter at a target venue. Three failure modes are common:

- **Ungrounded criticism:** weaknesses that the paper already answers, or locations and numbers that do not exist.
- **Compressed scores:** every paper rated "borderline".
- **Leakage:** the reviewer finds the paper's own camera-ready version, reviews or acceptance status, through retrieval or memory.

Prior AI reviewers either prompt a general model once per paper, as in the AI Scientist [4], or fine-tune an open model on public reviews: OpenReviewer [2], SEA-E [8], CycleReviewer [7] and DeepReviewer [12]. Academic Paper Reviewer instead addresses these failure modes with structure rather than with a single prompt or a single trained model.

- Several reviewers with distinct roles write independently.
- A Devil's Advocate attacks the paper.
- A report check removes ungrounded points.
- An EIC acts as area chair.

The skill is tuned for computer vision, deep learning and ML papers, with task-specific checklists for detection, segmentation, 3D/NeRF/3DGS, restoration, generation, VLMs, video, medical imaging and efficient models, but it can be used in any field. Its output is a *simulation* to help authors improve a paper, not a prediction of a venue's decision.

Contributions.

1. A review workflow with independent role-based reviewers, a quality-weighted EIC, and deterministic score aggregation into venue-specific forms (Section 2).
2. Safeguards against prompt injection and self-leakage in literature search (Section 3).
3. A reproducible calibration harness following the paperreview.ai protocol [5], extended with CSPaper-style error metrics [6] and a no-split pilot mode (Section 5).
4. An evaluation on 86 ICLR 2026 papers of both the ratings (Section 5) and the stated weaknesses against the human reviews, including whether the important issues and the EIC's priorities match the humans' (Section 6); and the pilots that led there, including what did not work.

2 System design

2.1 Workflow

A full review runs the phases in Figure 1. The phases are Markdown instructions in `SKILL.md` and `agents/*.md`; scripts are optional helpers.

Phase 0: field analysis. A field analyst identifies the paper's sub-field, technique, paper type (empirical, theoretical, dataset, system, benchmark, analysis, position) and target venue, and configures five reviewers with specific expertise. An example is "token pruning + latency benchmarking" rather than "deep learning". `integrity_scan.py` flags text addressed to reviewers or AI systems, as well as hidden characters.

Phase 1: independent reviews. Five reports are written independently: Methodology (R1), Domain (R2), Perspective (R3), Generalist (R4) and a Devil's Advocate (DA). In Claude Code they can run as parallel subagents. The Methodology reviewer runs an error audit in nine categories (result inconsistencies, equation and proof errors, assumptions, protocol flaws, claim distortion, inconsistent definitions, citation errors, omissions, figure/table integrity). It works from a shared factual *paper memory*, and every error needs a location, a verbatim quote and a concrete check.

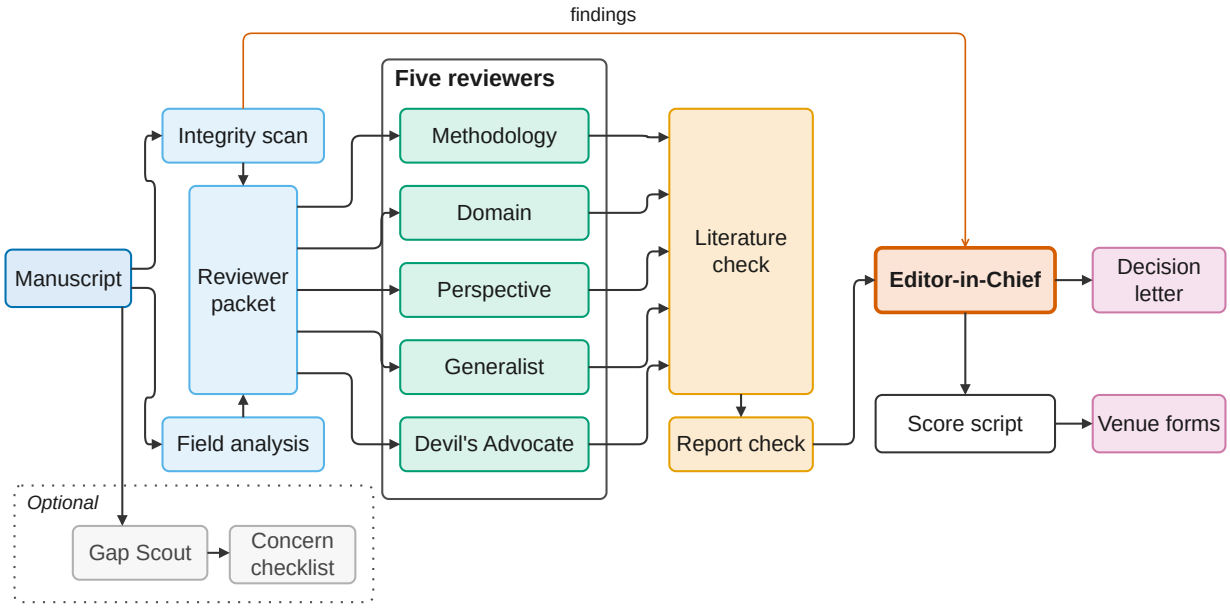


Figure 1: Full-mode workflow. Phase 0 analyses the field and configures the panel, and an integrity scan redacts text addressed to AI reviewers; each reviewer gets the configuration, a factual paper memory and the redacted copy, never another reviewer's report. The literature check (Phase 1.5, skipped without network access) cannot retrieve the paper under review; the report check (Phase 1.6) follows it. The Editor-in-Chief (Phase 2) alone sees the integrity findings (thin arrow), weights each review by quality and decides; a deterministic script translates the scores into the forms of 11 venues. The Gap Scout is optional, runs after the review and changes none of it. Drawn with the paper-figure skill (Draw.io, editable source in `report/fig/`).

Phase 1.5: literature check. This runs only with network access and after the reports are drafted. Searches are driven by contribution claims and benchmarks: a novelty verdict per claim, a baseline scout, a coverage pass, a field snapshot and a similarity ranking. Section 3 describes how the paper itself is excluded.

Phase 1.6: report check. Before the EIC reads anything, vague weaknesses, points the paper already answers, and claims the paper contradicts are removed. Cosmetic issues are grouped as Minor.

Phase 2: EIC meta-review. The EIC reads every report and checks the decisive points against the paper. It assigns each reviewer a quality weight in $[0.5, 1.5]$, states its own pool position, and decides. Each DA-CRITICAL issue gets an area-chair test ("would I reject on this alone?").

Other modes reuse the same components: quick assessment, methodology focus, guided (Socratic) review, rebuttal drafting and re-review of a revision.

Post-review analysis (optional). After the review, or on its own, the user can ask for a *Gap Scout*: a separate agent that reads only the redacted paper and lists 20–30 concrete concerns reviewers are likely to raise (the specific missing experiment, baseline, definition or setting, with a location). It gives no scores, is never called by default and never changes the review; its list is a checklist for the authors (Section 6).

2.2 Scores

Every non-DA reviewer gives NeurIPS-style 1–4 ratings, an Overall 1–6 recommendation (6 Strong Accept . . . 1 Strong Reject), a confidence 1–5, and seven 0–100 dimension scores: technical rigor, clarity, novelty, significance, experimental validation, reproducibility and ethics. The DA gives

Table 1: Paper-type weights of the seven 0–100 dimensions (each row sums to 1).

Type	Rigor	Clarity	Novelty	Signif.	Exp. val.	Repro.	Ethics
empirical	.20	.10	.15	.15	.20	.10	.10
theoretical	.30	.10	.20	.15	.10	.05	.10
dataset	.15	.10	.10	.20	.15	.15	.15
system	.20	.10	.10	.20	.20	.10	.10
benchmark	.15	.15	.10	.20	.20	.15	.05
analysis	.25	.10	.20	.20	.15	.05	.05
position	.15	.20	.25	.25	.05	.00	.10

the 0–100 set only. Dimensions are combined with paper-type weights (Table 1).

Two score lists are kept. The **official** list holds the reviewers' raw ratings and their plain means, which is what authors see; it is never modified. The **EIC assessment** weights reviewer i by

$$w_i = c(\text{conf}_i) \cdot q_i, \qquad c(5..1) = (1.00, 0.85, 0.70, 0.55, 0.40), \quad q_i \in [0.5, 1.5], \tag{1}$$

where q_i is the EIC's quality weight. The decision starts from the EIC-weighted mean of the Overall ratings, rounded half up. The weighted 0–100 composite is a cross-check: it maps to Overall at thresholds 85/70/60/50/35. Two further rules apply:

- **DA cap.** Upheld DA-CRITICAL issues cap the decision at 4 if they can be resolved by clarification and at 3 otherwise.
- **Consistency warnings** flag disagreements that the EIC must explain.

Calibration to the submission pool (2.10). Each reviewer and the EIC first state a *pool position* P: the share of the venue's submissions the paper is better than, using real base rates (about 25% acceptance at major ML/CV conferences; ICLR 2026 about 27% accepted, 45% rejected, 24% withdrawn, 5% desk rejected). They then score within the matching band:

P	≥ 95	85–94	70–84	55–69	30–54	< 30
composite	85+	73–84	62–72	55–61	42–54	< 42

A composite of 62 is the accept-side bar. Uncertainty is never a reason to score a paper as borderline. The aggregation script warns when a composite lies more than 5 points outside its pool band, and when all reviewers sit in the borderline band.

2.3 Venue forms

Internal scores are translated into 11 venue forms: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAAI, ACM MM, TMLR, IEEE TPAMI and IEEE TIP. Each translation applies the venue's emphasis weights, bar offset and scale. Each scale is labeled *official*, *confirmed*, *secondary*, *assumed* or *unpublished*. For ICLR 2026 the Overall maps to the rating set $\{0, 2, 4, 6, 8, 10\}$ as $r = 2(x-1)$. The legacy (2025) set is $\{1, 3, 5, 6, 8, 10\}$.

3 Safeguards

- **Prompt injection.** The manuscript is treated as data, never as instructions. Injected text is reported to the editor as an integrity concern and never raises a score.
- **Self-leak in literature search.** A fingerprint filter refuses queries that could find the reviewed paper. It drops the paper's own arXiv versions, code pages, reviews and acceptance lists unread, and labels every other result as prior (peer-reviewed or 6+ months old), recent preprint, concurrent or newer. Only prior work can be a required comparison. Each run logs the mode (`on`, `strict` or `off`). In `strict` mode a cutoff date applies; in evaluation this is the submission deadline.

Table 2: Grounding on 30 accepted papers: share of cited items not found in the paper. Numbers the reviewers recomputed themselves (525) are excluded and were spot-checked by hand.

Check	Checked	Not found	Rate
Cited locations (tables, figures, sections, equations)	1101	48	4%
Quoted phrases	778	9	1%
Quoted numbers	842	2	0%

- **Recognition.** Reviewers report when they recognize a paper from prior knowledge, and such papers are reported separately.
- **Integrity.** Reviewer personas are fictional, no citations may be fabricated, and every point is tied to a location in the manuscript.

4 Evaluation I: accepted papers and planted flaws

An earlier release of the skill (before 2.10) reviewed 30 recently accepted papers: 10 each from CVPR, ECCV and ICLR, spread over 10 topics. All runs completed with every required section; the median review length was 10,214 words. Table 2 reports grounding.

Planted flaws. In six accepted papers we inserted one sentence admitting a fatal evaluation flaw: hyperparameters tuned on the test set, or the test split merged into training.

- The reviews mentioned the flaw in 6/6 papers.
- The Devil's Advocate raised it as DA-CRITICAL in 2/6, and the EIC upheld both.
- The final decision was ≤ 3 in 5/6.

The miss was an ICLR paper rated 4.

Decisions. 16/30 accepted papers landed on the accept side (Overall ≥ 4); 14 received 3. This compression motivated the pool-position calibration of version 2.10. The DA raised CRITICAL issues on 11/30 papers, and the EIC upheld none on real papers.

5 Evaluation II: ICLR 2026 ratings

5.1 Protocol

We follow the paperreview.ai protocol [5].

- Sample ICLR 2026 submissions with human reviews, including withdrawn ones.
- Fit an ordinary least-squares map from the seven dimension scores to the human rating on a train split.
- Report on the test split: Spearman with one human rating, and the AUC of the score for acceptance. Confidence intervals come from bootstrap resampling; results are given for all scored papers and without withdrawn papers.

The harness (`evals/iclr_calibration.py`, standard library only) runs in GitHub Actions through the API. Reviewers never see human scores or decisions.

Error metrics. Following the CSPaper benchmark [6], for predictions $\hat{s}_j$ and human ratings s_j on a scale $[S_{\min}, S_{\max}]$ (ICLR 2026: $[0, 10]$),

$$\mathrm{NMAE} = \frac{1}{n} \sum_j \frac{|s_j - \hat{s}_j|}{S_{\max} - S_{\min}}, \qquad \mathrm{NME} = \frac{1}{n} \sum_j \frac{\hat{s}_j - s_j}{S_{\max} - S_{\min}}. \tag{2}$$

Table 3: Full evaluation (Sonnet, high effort, 86 papers): the panel rating against the mean human rating, Spearman [bootstrap 95% CI] / acceptance AUC, and errors on the ICLR scale.

Papers	n	Spearman	AUC
All	86	+**0.47** [0.27, 0.64]	0.64
Pilot 3's papers	20	+0.48 [0.10, 0.74]	0.67
The 66 new papers	66	+0.48 [0.25, 0.66]	0.64
Not recognized by the reviewer	75	+0.48 [0.28, 0.65]	0.66
Without withdrawn papers	69	+0.40 [0.17, 0.59]	0.61
Test split, against one human (protocol)	41	+0.45 [0.14, 0.69]	0.67
Protocol regression on 7 dimensions, test split	41	+0.19 [−0.13, 0.48]	0.67
References on the same 86 papers: one human vs another +0.14, one-human AUC 0.64.			
Error (86 papers)		NMAE ↓	NME
Panel rating		**0.084** ± 0.063 [0.071, 0.098]	−0.003 ± 0.105
Final rating (integer)		0.087 ± 0.064 [0.074, 0.102]	−0.018 ± 0.107
Constant at the median human mean		0.089 ± 0.074	–
One human vs mean human		0.118 ± 0.087	–
One human vs another human		0.188 ± 0.157	–

We report each with its per-paper standard deviation and a bootstrap 95% CI. Pilots are too small to fit the regression, so the `pilot` command scores the skill's own outputs as they are, without any fitting to human scores:

- the *panel rating*: the reviewers' mean Overall, mapped to the ICLR scale by $2(x-1)$;
- the *final rating*: the integer decision on the ICLR scale.

5.2 Data and a decision leak

The sample is ICLR 2026 with seed 2025: 300 papers drawn, 295 with human scores. Our first pilot used the PDFs from paperreview.ai's dump. These turned out to leak the decision: every accepted paper was a camera-ready version (author names, acknowledgments, no line numbers), while all other papers were anonymous submissions. Overall, 0/29 accepted papers and 56/56 other papers showed "Anonymous authors". We therefore use only arXiv versions posted before the 2026-01-26 decision date, normalized identically. The header before the abstract is replaced by the title, and line numbers, arXiv stamps, emails, affiliation footnotes, code links, acknowledgments and acceptance marks are removed. The literature check runs in `strict` mode with the submission deadline as cutoff.

The protocol's balanced set has 86 papers (23 accepted, 43 rejected, 17 withdrawn, 3 desk rejected; 45 train, 41 test). The pilots used 20 of them, stratified by decision and human score: 5 accepted, 10 rejected, 4 withdrawn and 1 desk rejected.

5.3 Main result: 86 papers

On the balanced set we reviewed the 66 papers not in the pilot with skill 2.10.1, which keeps the 2.10 panel, scoring and decision, and reused the 20 pilot 3 runs. All 86 runs finished, at $166.61 in total ($1.94 per paper). Table 3 gives the results.

- **The pilot signal holds.** The 66 new papers rank exactly as well as the 20 pilot papers (0.48), and recognized papers do not drive the result: the reviewer reported knowing 11 of the 86, and without them Spearman is still 0.48.
- **Above human–human agreement.** The panel ranks papers far more like the human mean than one human ranks like another (0.47 vs 0.14); on the test split against one human it reaches 0.45, roughly the level paperreview.ai reports for its AI reviewer (0.42, on other papers from another year). Acceptance AUC (0.64) equals one human's here.

Table 4: Ranking on the 20-paper pilot: Spearman with the mean human rating [bootstrap 95% CI] / acceptance AUC. Human references are on the same papers unless noted.

Run	Panel rating (1–6)	EIC pool position	Mean of 6 dimensions
Haiku, default	+0.20 [−0.21, 0.58] / 0.56	−0.06 [−0.55, 0.42] / 0.42	−0.01 [−0.52, 0.49] / 0.49
Sonnet, high	**+0.48** [0.09, 0.74] / **0.67**	+0.30 [−0.14, 0.69] / 0.55	+0.22 [−0.27, 0.64] / 0.57
Without the 3 papers Sonnet recognized ($n = 17$):			
Haiku, default	+0.07 / 0.50	−0.25 / 0.33	−0.10 / 0.44
Sonnet, high	+0.46 [0.09, 0.74] / 0.69	+0.22 / 0.53	+0.10 / 0.54
References: one human vs another +0.12, one-human AUC 0.58 (same papers); human vs human on our ICLR 2026 test split 0.25 / 0.75.			

- **The fitted regression does worse than no fitting.** The protocol's least-squares map from seven dimensions, fitted on 45 papers, reaches only 0.19 on the test split; three of its seven weights are negative. The unfitted panel rating is the better score.
- **No stage loses signal.** The EIC-weighted rating (0.45), the reviewers' pool position (0.48) and the panel without the Devil's Advocate (0.48) are all within ±0.02 of the panel rating, with intervals that include zero.
- **Compression.** The final Overall is 3 for 67 of 86 papers. The mean panel rating (ICLR scale) is 4.45 for accepted papers against a human 5.28, and 4.10 for rejected papers against 3.89: the order is right, but the gap is 0.35 points instead of 1.39.
- **Withdrawn papers help.** Without them Spearman drops to 0.40; part of the signal is spotting clearly weak submissions.
- **No reweighting or rescaling of the panel's numbers does better.** Chosen on the 45 train papers and checked on the 41 test papers, the best alternative is the mean Novelty score (+0.53 on all papers, +0.51 on test), but its gain over the panel rating, +0.06 [−0.10, +0.22] on test, is within noise. Clarity, Reproducibility and Ethics barely rank the papers (+0.15 to +0.25). A linear stretch of the panel rating fitted on train has slope 0.79, not above 1, and lowers the test NMAE only from 0.086 to 0.081: at this correlation the best predictor is more compressed, not less. What is left to improve is in the review content.

The reviewer's workspace contained no human scores for the 66 new papers; the workflow now also removes all evaluation data from it before each review.

5.4 Pilots and model choice

Five runs led to the evaluated version; Tables 4 and 5 give the 20-paper pilots of Haiku and Sonnet.

Pilot	Skill	Model / effort	Outcome
1	2.9	Haiku / default	decision leak in the text; scores compressed; Spearman −0.12, AUC 0.47
2	2.10	Haiku / default	normalized text; Spearman −0.01, AUC 0.49 (mean of 6 dimensions)
3	2.10	Sonnet / high	normalized text; Tables 4–5
4	2.11	Sonnet / high	redesigned panel; Spearman 0.32, AUC 0.55; withdrawn (Section 6.5)
Full	2.10.1	Sonnet / high	balanced set, 86 papers; Section 5.3

In pilot 1 nearly every paper received Overall 3, including papers with human means of 2.0 and 6.0. The anchors described papers only relative to *accepted* papers, and the anti-harshness rule could only raise scores. Version 2.10 added pool positions, base rates, two-sided checks and EIC rulings on borderline panels.

What the pilots showed.

Table 5: Error on the ICLR 2026 scale (0–10) against the mean human rating, 20 papers. NMAE ± std [95% CI]; NME ± std.

Predictor	NMAE ↓	NME	Spearman	AUC
Haiku, panel rating	0.093 ± 0.057 [0.068, 0.118]	$+0.030 \pm 0.105$	+0.21	0.56
Haiku, final rating	0.085 ± 0.063 [0.058, 0.115]	-0.020 ± 0.104	n/a (constant)	0.50
Sonnet, panel rating	$\mathbf{0.079} \pm 0.061$ [0.054, 0.106]	$\mathbf{+0.004} \pm 0.100$	**+0.48**	**0.67**
Sonnet, final rating	0.090 ± 0.064 [0.061, 0.119]	$+0.010 \pm 0.110$	+0.26	0.53
Constant at the median human mean	0.085 ± 0.060	–	–	–
One human vs another human	0.180 ± 0.125	–	+0.12	–
One human vs mean human	0.102 ± 0.065	–	–	–

- **Model capability matters.** Haiku's scores did not separate papers. Its integer final rating was the same for all 20 papers, so Spearman is undefined. Sonnet at high effort gave the first signal whose CI excludes zero, at about the level of human–human agreement.
- **NMAE rewards the middle.** An oracle constant at the median human mean scores 0.085. That beats one human against another (0.180), and Haiku's constant output scores the same. NMAE must therefore be read together with a ranking metric. Only Sonnet's panel rating beats the constant *and* ranks papers.
- **No bias after calibration.** NME is close to zero for both models, so the 2.10 bands no longer shift scores up or down on this pool.
- **The integer decision loses signal.** Rounding the panel mean to an integer decision lowers Spearman from 0.48 to 0.26 for Sonnet. For ranking, the continuous panel rating should be the primary score.
- **Recognition.** The three papers Sonnet recognized all scored high (EIC pool position 65–70) regardless of outcome. Removing them changes the Sonnet results little.
- **Cost.** About $0.86 per paper for Haiku and $1.97 per paper for Sonnet at high effort.

For reference only: CSPaper reports NMAE 0.128, NME −0.015 and SRC 0.658 for Claude Sonnet 5.5 on ICLR main. That benchmark's sample size and paper overlap are unknown, so the numbers are not directly comparable with ours.

5.5 Other model families

The skill is model-neutral Markdown, so we ran version 2.10.1 unchanged on pilot 3's 20 papers with two OpenAI models through the Codex command-line tool, at high reasoning effort, with the same prompt and blinded text. All 40 reviews finished.

Table 6: Three models on the same 20 papers. Spearman with the mean human rating [bootstrap 95% CI] and acceptance AUC for the panel rating; NMAE and NME for the ICLR rating. GPT costs are estimates from token counts at list prices; the Terra estimate priced cached input as fresh input and so overstates its cost.

Model, high effort	Spearman	AUC	NMAE / NME	Cost per paper
Sonnet	**+0.48** [0.12, 0.73]	0.67	0.090 / +0.010	$1.97
GPT-6.1 Sol	+0.02 [−0.38, 0.45]	0.47	0.126 / −0.020	≈$1.39
GPT-6.1 Terra	−0.01 [−0.43, 0.43]	0.40	0.133 / −0.070	≈$4.55

Neither GPT model ranks the papers (Table 6), and no stage of the pipeline helps: the reviewers' and the EIC's pool positions are just as flat. Both do worse than a constant guess on NMAE (0.085). The failure is the same for both: Sol gives the three papers with the highest human mean (6.0) 3.4, 2.8 and 2.8 on the 1–6 scale, and the two with the lowest 3.2 each. Terra is also harsher, with six papers at ICLR 2 where Sonnet has none. The models agree with each other more than with the humans: the panel ratings correlate at 0.71 (Sol–Terra), 0.60 (Sol–Sonnet)

and 0.65 (Terra–Sonnet). With 20 papers the marginal intervals overlap, so this is evidence that the result depends on the model, not a ranking of the three; the paired test below is tighter.

Ranking is not the only measure. We repeated the issue-overlap study of Section 6 on the GPT reviews of the same 20 papers (Table 7). The GPT models raise about half as many main issues per paper as Sonnet, and a similar share of what they raise is something a human also raised, so their reviews are shorter at the same precision. Sol is also the cheapest of the three and unbiased on average, both GPT models use more of the rating scale (ICLR 2 / 4 / 6: Sol 3 / 14 / 3, Terra 6 / 13 / 1, Sonnet 0 / 17 / 3), and they recognized fewer papers (1 and 0 against 3). They fall short on coverage: they raise half as many of the humans' issues, and the gap is largest on the issues that two or more humans agree on. The count uses each report's weaknesses section only; Sol's reviews are as long as Sonnet's but put more findings in separate audit sections, which the study leaves out for every model.

Table 7: Issue overlap with the human reviewers, three models, same 20 papers, main issues only [bootstrap 95% CI].

	Sonnet	GPT-6.1 Sol	GPT-6.1 Terra
Human issues raised by the panel	**0.45** [0.37, 0.54]	0.21 [0.17, 0.25]	0.27 [0.21, 0.34]
raised by 2+ humans	**0.78**	0.27	0.50
AI issues also raised by a human	0.26 [0.22, 0.29]	0.23 [0.19, 0.28]	**0.28** [0.23, 0.34]
AI main issues per paper	20.9	10.8	10.9

Why Sol does not rank the papers. Sol reads the whole paper, searches more than Sonnet and writes reviews of the same length, so its flat ranking was unexpected. A paired bootstrap (5000 resamples) of Sonnet's minus Sol's panel-rating Spearman gives 0.46 [0.08, 0.81], $P(\leq 0) = 0.01$: the gap is real, its size uncertain. Sol is not noisy either (its reviewers' pool correlates 0.81 with Terra's and 0.61 with Sonnet's); it ranks by another standard. On the humans' top ten papers its mean pool position is 55.1, on the bottom ten 55.3 (Sonnet: 51.4 against 46.6). Three mechanisms show in the data (Table 8):

- **Its scores follow rigor, the humans' follow novelty.** Sol's pool is driven most by technical rigor (Spearman 0.94 with its own pool), a dimension that does not track the humans (+0.04); Sonnet's is driven most by novelty (0.92), which does (+0.56).
- **Lenient where the problem is novelty.** On four of the humans' weakest papers Sol's pool sits 12–16 points above Sonnet's. The human reviews of these papers lead with lack of novelty and name the prior work; none of Sol's top concerns on any of the 20 papers cites a prior work, against 6 of 20 papers for Sonnet. Search volume is not the cause (median 190 search-ledger entries per paper against Sonnet's 129): the searches do not surface the overlapping paper, so the evidence rules leave the doubt unproven and lightly weighted.
- **Harsh where a proof defect is verifiable.** A theory paper with human mean 5.0 is Sonnet's 7th of 20 (ICLR 4) and Sol's 19th (rigor 38.5, ICLR 2) after Sol logged a series of proof errors as verified; Sonnet found the same main defect and still gave ICLR 4. An accepted theory paper with human mean 6.0 is Sol's last.

Sol also reasons less: median 9.3k reasoning tokens per paper against Sonnet's 30.2k (Terra 4.6k), although it runs longer (14.0 against 9.6 minutes); the time goes to tool calls. Reading, review length, search volume and reviewer independence are ruled out. With 20 papers and keyword categories this is a diagnosis, not a proof, and agreeing with the humans is not the same as being right: Sol's proof findings may be correct. Untested fixes: require a named closest prior work (or an explicit "none found") behind every novelty concern, grade verified errors as fatal or repairable before they move the score, and rerun Sol at a higher reasoning effort.

Table 8: Why Sol does not rank pilot 3's 20 papers. Spearman of a dimension with the model's own reviewers' pool / with the human mean; human issues hit, by keyword category (crude); medians per paper.

	Sonnet	GPT-6.1 Sol	GPT-6.1 Terra
Technical rigor: own pool / human	0.84 / +0.23	**0.94** / +0.04	0.91 / −0.11
Novelty: own pool / human	**0.92** / +**0.56**	0.76 / +0.19	0.61 / +0.32
Significance: own pool / human	0.82 / +0.47	0.78 / +0.31	0.91 / +0.14
Pool, human top 10 / bottom 10	51.4 / 46.6	55.1 / 55.3	50.2 / 52.3
Human novelty issues hit	**0.70**	0.38	0.44
Human experiment issues hit	**0.49**	0.18	0.23
Top concerns naming a prior work (papers)	6 / 20	0 / 20	0 / 20
Reasoning tokens	30.2k	9.3k	4.6k
Wall time (min)	9.6	14.0	8.5

6 Evaluation III: weaknesses compared with the human reviews

Scores say nothing about *which* weaknesses a review raises. On the 86 papers of the full evaluation (Sonnet, high effort, skill 2.10.1) we compared the weakness sections of the AI panel with those of the ICLR 2026 human reviews (3–5 per paper). For each paper, one Sonnet annotator saw both panels under random names, in random order and without roles. It split them into atomic points, grouped the points into issues within each panel and matched issues across panels; it did not judge correctness and never saw scores or decisions. We count main issues only (minor and cosmetic points excluded) and give 95% bootstrap intervals over papers. The method was set on pilot 3's 20 papers, where four papers annotated twice by independent annotators differ by at most 0.04 on every metric; every pilot hit rate is reproduced within 0.06 on the 66 new papers. The tool and the annotation prompts are in the repository (`evals/issue_overlap.py`).

6.1 Coverage

Table 9: Issue overlap and hit rate, 86 ICLR 2026 papers, main issues (pilot 3's 20 papers in the last column).

	Human	AI, 5 reviewers	AI, R1–R4
Main issues per paper	11.8 [11.1, 12.6]	20.4 [19.5, 21.2]	–
One reviewer only	0.77 [0.74, 0.80]	0.48 [0.46, 0.50]	0.73 [0.71, 0.76]
Pairwise Jaccard	0.11 [0.09, 0.13]	0.15 [0.14, 0.16]	0.09 [0.08, 0.10]
Hit rate	86 papers		Pilot (20)
Human issues hit by the AI panel	**0.46** [0.42, 0.49]		0.45
raised by 2+ humans	**0.74** [0.67, 0.80]		0.78
raised by one human	0.38 [0.34, 0.42]		0.38
AI issues also raised by humans	0.26 [0.24, 0.28]		0.26
One human's issues, by other humans	0.41 [0.36, 0.45]		0.39
One human's issues, by 1 / 3 / 5 AI reviewers	0.21 / 0.42 / 0.52		0.20 / 0.40 / 0.50

- **Redundancy comes from two roles.** The five-reviewer panel repeats itself more than a human panel, but at equal size (R1–R4) its overlap matches the humans' (Jaccard 0.09 vs 0.11). The Devil's Advocate raises 93% of its issues together with another role, and R4 79%. Two thirds of the Devil's Advocate's issues are also R1's (and 61% of R1's are also the Devil's Advocate's), and half of R4's are also R1's or the Devil's Advocate's; every pair among R1, R2 and R3 shares at most a fifth. The prompts explain it: the Devil's Advocate runs "the same full-text search as the Methodology Reviewer", recomputes the headline numbers R1's

Table 10: Per-role contribution, 86 papers (means per paper). The last two columns are the panel of the other four: its hit rate (full panel 0.46) and the Spearman of its mean Overall with the mean human rating (full panel +0.47; every difference has a 95% CI that includes zero).

Role	Issues	Raised by no other role	Human issues hit	Hit by no other role	Without it hit rate	Without it Spearman
R1 Methodology	9.9	0.25	0.22	0.02	0.44	+0.48
R2 Domain	5.4	0.51	0.18	**0.08**	**0.38**	+0.45
R3 Perspective	5.2	0.58	0.15	**0.06**	**0.40**	+0.46
R4 Generalist	4.8	0.21	0.14	0.02	0.44	+0.46
Devil's Advocate	8.9	0.07	0.21	0.01	0.45	+0.48

error audit recomputes, and checks baselines, alternative explanations and overclaims that R1's fairness, validity and claim-to-evidence steps already cover.

- **R2 and R3 carry what is unique.** They rarely repeat the others, and removing either costs the most hits (0.46 to 0.38 and 0.40); removing R1, R4 or the Devil's Advocate costs at most 0.02. No single role changes the ranking distinguishably (0.45–0.48).
- **The Devil's Advocate's value is in the decision, not the issue list.** Only 0.6% of the human issues are hit by it alone, and its own issues match a human issue 4% of the time. Its CRITICAL findings cap the decision: it raised one on 15 of the 86 papers, the EIC upheld 12, and on 2 papers the cap lowered the decision below the starting point, both toward the humans (a rejected and a withdrawn paper). Whether its repeated checks catch defects that R1 misses cannot be measured against human reviews, since a defect no human raised does not count; a seeded-defect check with planted, known defects is prepared for that. No role is removed or trimmed on this evidence; a narrower Devil's Advocate would first have to show, on the same seeds, that it loses nothing.
- **The panel finds what humans agree on.** It raises 46% of the humans' main issues and 74% of those two or more humans share. Three AI reviewers anticipate a given human's issues as well as the other humans do (0.42 vs 0.41). Coverage grows with panel size (0.18, 0.30, 0.37, 0.42, 0.46 for 1–5 reviewers).
- **Too many issues.** The panel raises 1.7× as many main issues, and a quarter are also raised by a human. Unmatched issues are not necessarily wrong, but the reader needs a short list.
- **Votes and issues differ.** On the pilot, R4 added few issues, yet the Spearman of the mean Overall dropped without it (R1–R4 0.52, R1–R3 0.48). Version 2.11 tested a panel that keeps R4's vote but limits its weaknesses, moves the Devil's Advocate into an independent Report Checker and adds 3–5 key concerns (Section 6.5).

6.2 Important issues

The human reviews carry no severity labels, so we use three proxies: how many humans raised the issue, the rating of the reviewers who raised it (on the ICLR 2026 scale, 4 or less is a reject vote) and whether it was among the first two main points of a review (Table 11). Every proxy points the same way: the more important the issue, the more often the panel raises it, from 0.29 for a single accept-voter's later point to 0.74 for consensus issues. Of the hit issues, 63% of the consensus ones and 56% of the others match an AI issue raised by two or more AI reviewers, so the panel's own agreement tracks importance only weakly. By decision, the hit rate is lower on accepted papers (0.39 [0.32, 0.45], 23 papers) than on rejected ones (0.49 [0.45, 0.54], 43 papers): their reviewers raise fewer issues (11.0 per paper), and fewer of the AI's issues match theirs (0.19 vs 0.29).

Table 11: Hit rate of the 1018 human main issues by importance proxy, 86 papers. "Raised by" counts main points only; the one issue the AI raised only as a minor point counts as not hit.

Human main issues	n	Hit rate [95% CI]
Raised by 2+ humans	219	**0.74** [0.67, 0.80]
Raised by one human	799	0.37 [0.34, 0.40]
Raised by a reviewer who voted reject (≤ 4)	790	0.48 [0.45, 0.52]
Raised only by reviewers who voted accept (≥ 6)	228	0.34 [0.28, 0.40]
Raised by the paper's lowest-rating reviewer	555	0.50 [0.46, 0.54]
Among a reviewer's first two main points	535	0.51 [0.47, 0.56]
Not among any reviewer's first two	483	0.38 [0.34, 0.42]
One human, accept votes only, not a lead point	91	0.29 [0.20, 0.38]

6.3 The EIC's priority list

Each final report ends with the EIC's five top concerns, the first thing an author reads. To see whether they are the humans' priorities, one Sonnet annotator per paper saw the human main issues and, mixed at random without sources, the five top concerns and the panel's main issues, and matched each human issue to the items that raise the same specific point. The re-judged panel hits agree with the original annotation on 85% of human issues.

Table 12: The EIC's five top concerns against the human main issues, 86 papers. Left: share of AI items that match a human issue. Right: of the human issues the panel raised, the share that also appear among the top five.

AI item matches	Top concern	Panel issue	Human issue raised by the panel	n	In top 5
A human issue	**0.34**	0.26	2+ humans	161	**0.45**
A consensus issue	**0.16**	0.10	One human	305	0.30
A reject-vote issue	0.29	0.22	Behind a reject vote	384	0.37
Match or related	0.65	0.55	Accept votes only	82	0.29
Items	430	1751	A reviewer's lead point	284	0.38

The answer is *partly* (Table 12). A top concern is 1.3× as likely as an average panel issue to match a human issue and 1.6× as likely to match a consensus issue, and the EIC promotes the consensus issues the panel found more often than single-reviewer ones (0.45 vs 0.30). But most human priorities do not reach the list: the five concerns match 34% of the consensus issues, against 74% for the whole panel (55% when related items count), partly because each concern bundles several points into one line. Within the five there is no order: the first matches a human issue 0.37 of the time and the fifth 0.29, with no trend in between. Two-thirds of the top concerns match no human issue exactly; they need not be wrong, but a priority list that leads with the humans' consensus issues would serve authors better.

6.4 What the panel misses

Of the 1018 human main issues, 456 were hit, 460 had only a related AI issue and 101 had nothing close. A Sonnet labeler (one per five papers) gave each issue a topic and the evidence needed to raise it (before seeing which were hit) and, for the 561 non-hits, the likely reason, by comparing it with the AI issue summaries (Table 13). 82% of the non-hits have a related AI issue: 244 are issues the panel raised only as a broad category ("more baselines") while the human named the item (a specific method or dataset), and 110 more are related concerns with a different diagnosis. True misses are mostly subjective scope requests (58) and fine technical detail (21). Method clarity (0.22), presentation (0.20) and scope (0.33) are hit least; missing experiments, the largest topic (28% of human issues), are hit 0.47. Issues that need outside knowledge are hit *more* often

than text-only ones (0.51 vs 0.44), so the literature check is not the bottleneck; figure-dependent issues are hit least (0.29) but are rare (45).

Table 13: Hit rate of the 1018 human main issues by topic and evidence needed, and the likely reason for the 561 non-hits (related: a partial AI overlap; missed: nothing close), 86 papers.

Topic	n	Hit	Reason (non-hits)	Related	Missed
Missing experiments	287	0.47	Raised only as a broad category	242	2
Scope, limitations	148	0.33	Different diagnosis	107	3
Method clarity	107	0.22	Subjective request	59	58
Novelty, related work	100	0.60	Fine technical detail	19	21
Baselines	99	0.63	Needs a specific prior work	14	8
Unsupported claims	91	0.37	Needs figures	11	6
Theory	81	0.51	Unclear	8	3
Efficiency	50	0.64			
Reproducibility	28	0.43			
Presentation	20	0.20			
Evidence needed					
Text and tables	721	0.44			
Outside knowledge	200	0.51			
Derivation	52	0.40			
Figures	45	0.29			

6.5 Pilot experiments: a Gap Scout and version 2.11

Both experiments ran on the 20 pilot papers only (247 human main issues).

A Gap Scout. Since most non-hits were raised only as a category, we tested one extra, unscored pass: a Sonnet agent that reads only the redacted paper and lists up to 30 concrete weaknesses reviewers would likely raise (claims and their missing evidence, named baselines, undefined elements, theory, scope, typical requests). One annotator per paper then matched each human main issue against the panel's issues and the Scout's items, mixed without sources, so both are judged on the same scale; the panel's re-judged hits agree with the original annotation on 88% of human issues. Table 14 gives the results. Twenty concrete items hit as many human issues as the whole panel; thirty hit 1.4× as many, and together with the panel 73%. The gain is largest for missing experiments (0.35 → 0.76) and baselines (0.55 → 0.82). Part of it is volume (about 51 items per paper against 12 human issues), and the Scout's items were not checked against the paper in this test. Version 2.10.1 offers the Scout as an optional post-review checklist of anticipated reviewer concerns, outside the review (Section 2).

Table 14: Human main issues hit (247 issues, 20 papers), judged by one annotator per paper.

Source	Items/paper	Issues hit	Hit rate [95% CI]
Five-reviewer panel	20.9	110	0.47 [0.38, 0.55]
Gap Scout, first 10 items	10	70	0.30 [0.23, 0.39]
Gap Scout, first 20 items	20	108	0.46 [0.39, 0.54]
Gap Scout, all	30	153	**0.65** [0.59, 0.71]
Panel + Gap Scout	50.9	175	**0.73** [0.68, 0.79]

Pilot 4: building it into the panel (version 2.11). Version 2.11 put the findings above into the review itself: four voting reviewers (R1 and R2 asked to name concrete items, R4 limited to presentation and venue fit), the Gap Scout as a sixth, unscored pass, and a fresh-context

Report Checker that ran the literature and report checks, the Devil's Advocate challenge, the key concerns, base reviewer weights and its own independent estimate. We ran it on the same 20 papers (Sonnet, high effort) and annotated hits on the same 247 human main issues, with one fresh annotator per paper (Table 15).

Table 15: Pilot 4 (skill 2.11) against pilot 3 (skill 2.10), same 20 papers. Hit rate of the 247 human main issues [95% CI]; ranking against the mean human rating.

	Pilot 3 (2.10)	Pilot 4 (2.11)
Panel hit rate	0.47 (five reviewers)	0.43 [0.35, 0.53] (four)
Gap Scout hit rate	0.65	0.65 [0.58, 0.72]
Panel + Gap Scout	0.73	0.74 [0.68, 0.80]
Devil's Advocate hit rate	–	0.01 (in the checker)
Panel rating: Spearman / AUC	**0.48** / **0.67**	0.32 [−0.19, 0.74] / 0.55
Panel rating: NMAE / NME	**0.079** / +0.004	0.083 / −0.028
Final integer rating: Spearman / AUC	0.26 / 0.53	0.24 / 0.47
Checker's own estimate: Spearman / AUC	–	0.51 [0.15, 0.77] / 0.67
Cost per paper	$1.97	$6.93

- **No gain in hits.** The named-item instructions did not raise the panel's hit rate; the additional hits still came from the Gap Scout (75 human issues no other source raised).
- **Worse ranking, compressed ratings.** R1 gave 2 on 13 papers and R4 gave 3 on 17; the paired difference of the panel rating's Spearman, pilot 4 − pilot 3, is −0.15 [−0.48, 0.17]. A plausible reading is that the more detailed instructions made the reports finer but their scores less aligned with the human ones.
- **The checker's estimate** ranked at 0.51, the only interval in pilot 4 that excludes zero, but it used three Overall values and is no better than the 2.10 panel's 0.48 within the error bars (difference to the 2.11 panel 0.19 [−0.32, 0.70]).

Version 2.10.1 therefore returns the panel, its instructions, the scoring and the decision rule to 2.10, keeps the Gap Scout only as an optional post-review checklist, and removes the Report Checker.

Caveats: one LLM annotator and one labeler per paper; the Gap Scout test covers only the 20 pilot papers; importance is inferred from the human reviews, not judged; human reviewers saw the figures, the AI read the extracted text; agreement with humans is not correctness.

7 Evaluation IV: the same model without the skill

The evaluations above compare the skill with human reviewers. They do not say how much of the agreement comes from the skill and how much from the model. We therefore ran a control on the same 86 papers: one agent with the same model and effort (Sonnet, high), the same blinded text and the same evaluation rules (the paper is untrusted text; no web access), given one sentence, "Review this paper. Target venue: ICLR.", and the ICLR form (rating 0, 2, 4, 6, 8 or 10). The skill's files are removed from its workspace: there are no reviewer roles, no EIC, no rubric and no literature check. The measures and predictions were written down before the run. Each paper was run once; all 86 completed, for $17.93 in total ($0.21 and about one minute per paper, against about $2 and 8–9 minutes for the skill). The skill is represented by the reviewers' mean rating, not by the EIC's integer rating.

- **Scores.** Ranking, acceptance and error are the same; every difference interval includes zero, also on the 41 held-out papers (Spearman 0.45 and 0.51, AUC 0.69 and 0.65).
- **The control uses three ratings.** It gives 2, 4 and 6 to 8, 69 and 9 papers and never 0, 8 or 10, although the human reviewers of these papers gave 0 eight times and 8 twenty-three times (of 336 ratings). Its correlation comes from the 17 papers it moves off 4: none of the eight

Table 16: The skill and the same model without it, 86 ICLR 2026 papers [95% CI]. The last column is the paired-bootstrap interval of the difference. Issues: the 1018 human main issues, matched blind by one annotator per paper against the items of both reviews.

	Skill (panel)	Without the skill	Difference
Spearman, mean human rating	0.47 [0.28, 0.63]	0.45 [0.28, 0.60]	[−0.12, 0.16]
AUC, acceptance	0.64 [0.51, 0.77]	0.63 [0.54, 0.71]	[−0.09, 0.12]
Pairwise order correct	0.68	0.62	
Tied paper pairs	0.13	0.66	
NMAE / NME	0.084 / −0.003	0.085 / −0.013	
Human main issues raised	0.48 [0.45, 0.52]	0.46 [0.42, 0.50]	[−0.02, 0.06]
raised by 2+ humans (221)	0.71	0.68	
raised by one human (797)	0.40	0.38	
Own items matching a human issue	0.27 [0.25, 0.29]	0.28 [0.25, 0.31]	[−0.04, 0.01]
Items per paper	20.4	23.4	
Planted defects found (3 papers, 15)	15	12	
number defects (7)	7	4	
Score change with the defects planted	−7 to −10 of 100	none	
Cost per paper	about $2	$0.21	
Weakness points carrying (3054 and 2357 points, keyword rules)			
a location (§, table, figure, equation)	0.60 [0.58, 0.62]	0.26 [0.23, 0.30]	
a number from the paper	0.42 [0.40, 0.45]	0.25 [0.22, 0.28]	
a quotation of the paper	0.48 [0.46, 0.50]	0.33 [0.30, 0.36]	
a fix or missing experiment	0.54 [0.53, 0.56]	0.06 [0.05, 0.07]	
a named method or prior work	0.66 [0.64, 0.69]	0.48 [0.44, 0.51]	
Words per point (median)	49	27	

rated 2 was accepted (mean human rating 2.42) and five of the nine rated 6 were (5.00). The skill's final rating is the same judgement: it equals the control's on 82 of the 86 papers.

- **What the panel mean adds.** It is continuous (2.0–6.5 on the ICLR scale, SD 0.86; humans 0.5–7.5, SD 1.15), so few pairs tie. Within the 69 papers the control rates 4 it still orders papers slightly (Spearman 0.27 [0.02, 0.47]) but does not separate accepted papers (AUC 0.54 [0.38, 0.70]).
- **Review content.** A review of about 1,500 words raises as many of the humans' main issues as the five reports (about 11,000 words), with the same share of unmatched items. The two overlap on part of what they raise: 33% of the human issues are raised by both, 16% only by the skill and 13% only by the control. The annotator's new judgement of the panel agrees with the original annotation on 84% of the issues.
- **How concrete the points are.** The overlap asks whether an issue is raised, not whether an author can find and fix it. Keyword rules over every weakness point (`offline_signals.py specificity`) show the skill's points carry more on each count: a location in the paper (60% against 26%), the paper's own numbers (42% against 25%) and words (48% against 33%), a fix or missing experiment (54% against 6%) and a named method or prior work (66% against 48%). On a planted wrong average the skill recomputes it from the rows of another table (2.5 against the stated 1.5) and names both tables; the control does not flag it. The rules show that a point carries evidence, not that it is right, and the control's requests sit in its Questions section, which is not counted, so the fix row overstates the gap. A blind pairwise judgement of the issues both raise is the proper test and has not been run.
- **Planted defects.** Fifteen defects were planted in three papers (seven statements, seven numbers and one instruction to reviewers) and the reviews were read with the score lines hidden. The skill finds all 15 and the control 12. The three the control misses need a cross-check: a table average that is not the mean of its rows, a column that sums to 110%, and a

value that differs between two tables. Both flag the planted instruction and ignore it. The skill's severity is uneven: selection on the test set is graded critical on two papers and major on the third. With the defects planted the skill's EIC pool position falls on all three papers (42 to 33, 45 to 38, 55 to 45), while the control's rating stays at 4. Three papers and one run each show a direction, not a rate.

Taken together, the skill helps where the plain review is weakest. It raises the same human issues but states them so that an author can act: tied to a place in the paper, backed by the paper's numbers and with a fix. Its scores separate papers that the plain model rates alike, it catches errors that need cross-checking, and its score moves when such errors are present. On agreement with human reviewers the two are equal, and that measure counts only what the humans wrote: it does not judge the skill's error log, severity grades, revision roadmap or literature check. The skill's cost is about ten times higher. The next version builds on the measured gains: every weakness tied to a location, evidence and a fix, more checks with a verifiable answer (table arithmetic, text against tables, selection on the test set), evaluated with planted defects, at lower cost.

8 Related work

AI reviewer systems. Figure 2 places our system among published AI reviewers along two axes: how the model is obtained and how a review is produced.

- **Prompted general models.** The AI Scientist [4] reviews with one prompted pass, optionally with reflection and an ensemble.
- **Fine-tuned single pass.** OpenReviewer [2] fine-tunes Llama-3.1-8B on about 79k expert reviews; its ratings follow the human distribution, while general models rate too leniently.
- **Fine-tuned with a consistency check.** SEA [8] standardizes multiple reviews into one format (SEA-S), fine-tunes Mistral-7B on them (SEA-E), and regenerates a review when a mismatch score (SEA-A) finds it inconsistent with the paper.
- **Fine-tuned with simulated reviewers.** CycleReviewer [7] (8B–123B) simulates several reviewers and a senior reviewer and averages their scores. It is trained on ICLR 2024 reviews and serves as the reward model of CycleResearcher.
- **Fine-tuned with multi-stage reasoning.** DeepReviewer [12] (7B, 14B) is trained on DeepReview-13K, synthetic stepwise reviews of ICLR 2024–2025 papers, and reasons in stages with Fast, Standard and Best modes.

Our system is closest in spirit to CycleReviewer and DeepReviewer: a panel of perspectives and staged reasoning. It differs in four ways:

- it is a prompt-only skill for a strong general model, so it needs no training data and can use a newer model as soon as one is released;
- the reviewers have distinct roles, including an adversarial Devil's Advocate;
- an explicit EIC weights reviews by quality instead of averaging them;
- scoring is deterministic and venue-specific.

Reported scores of reviewer models. Table 17 collects published score-agreement numbers. They come from different paper sets, sizes and targets and are *not* comparable with our 20-paper pilot. Because all the fine-tuned models were trained on data up to ICLR 2024–2025, ICLR 2026 is unseen by them, which makes them fair open-weight baselines for our sample (future work).

Other studies report MAE 0.86 for OpenReviewer-8B and 1.10 for SEA-E under their own setup [3]. NAIPv2 [11] could not reproduce these systems and quotes their published numbers. Re^2 [10] evaluates all four fine-tuned systems on its review and rebuttal data.

Benchmarks.
paperreview.ai [5]. Score agreement and acceptance AUC on ICLR; our protocol follows it.

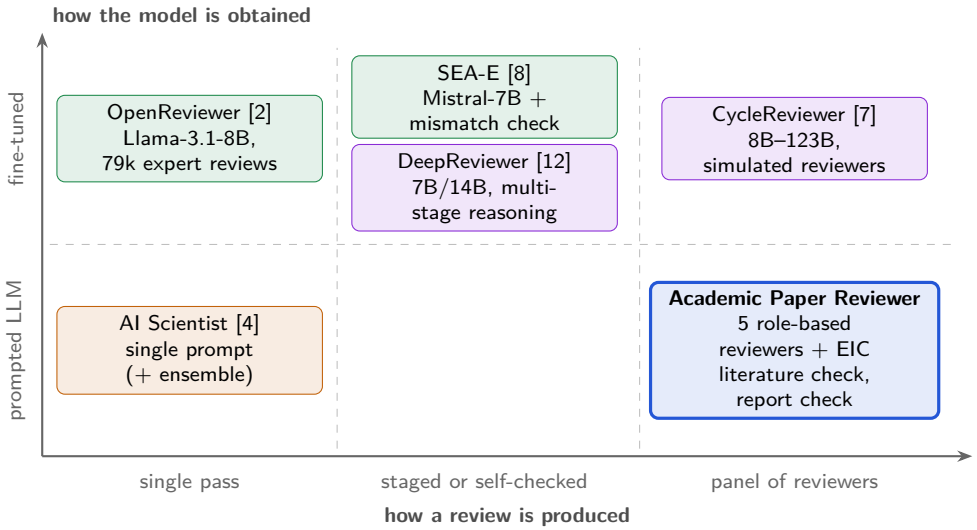


Figure 2: Design space of AI reviewers. Fine-tuned systems learn from public reviews of past conferences. Our skill prompts a strong general model through a structured panel and needs no training data. Positions are qualitative.

Table 17: Published rating agreement with the mean human rating (1–10 scale). Values are from Zhu et al. [12], Table 2 (columns ICLR 2024 / ICLR 2025). Last row: our pilot.

System	Spearman	Rating MAE	Decision acc.
DeepReviewer-14B [12]	0.356 / 0.405	0.91 / 0.92	0.64 / 0.69
CycleReviewer-70B [7]	0.336 / 0.267	1.25 / 1.21	0.63 / 0.68
CycleReviewer-8B [7]	0.280 / 0.279	1.24 / 1.21	0.64 / 0.68
AI Scientist, DeepSeek-R1 [4]	0.326 / 0.324	1.65 / 1.81	0.52 / 0.43
AI Scientist, Claude 3.5 Sonnet [4]	0.037 / −0.022	1.50 / 1.35	0.48 / 0.56
Ours, Sonnet high, panel rating (ICLR 2026, $n = 20$)	0.48	0.79†	–

† On the ICLR 2026 0–10 scale (NMAE 0.079 × 10).

CSPaper [6]. A score-prediction leaderboard with NMAE, NME, SRC and average word count; we adopt its error metrics.

CoCoReviewBench [1]. This benchmark scores review *content* rather than scores. It covers 3,900 ICLR 2017–2025 and NeurIPS 2021–2024 papers, with human references filtered for correctness through reviewer–reviewer and reviewer–author conflicts, using the meta-review as context. An LLM judge scores correctness, thoroughness, grounding, verifiability and clarity per category, and the benchmark also reports category completeness. Among its findings: lexical metrics reward overlap rather than correctness, reasoning models (including DeepReviewer-14B) match or beat humans on grounding and verifiability, and AI reviews are broad but shallow.

AI-text detection [9]. About 789k paired human and AI reviews for *detecting* AI-written reviews. It has a different goal and a restrictive license, so we do not use it.

9 Limitations and future work

- **Sample size.** The full evaluation has 86 papers; its Spearman interval is still 0.27–0.64, and the Gap Scout test and the 2.11 comparison use the 20-paper pilot only. Haiku and the two GPT-6.1 models were tested on the pilot's 20 papers only.
- **Gains over the plain model rest on little data.** Section 7 finds the same ranking,

acceptance AUC, hit rate and precision without the skill; the skill's gains (concrete points, score resolution, planted defects found, scores that react to them) are measured on one run per paper; concreteness is counted with keyword rules, not judged, and the planted-defect comparison covers three papers.

- **One venue and one year.** Calibration covers ICLR 2026 only. Other venues use assumed or secondary scales.
- **Memorization.** Strong models recognize some papers. We report recognized papers separately but cannot detect unreported recognition.
- **Scores are not the review.** Ranking agreement says nothing about whether the stated weaknesses are correct, and the overlap study (Section 6) measures agreement with humans, not correctness. A content-level evaluation (for example CoCoReviewBench [1]) is the natural next step, together with open-weight baselines [7, 12] run on the same ICLR 2026 papers.
- **The Gap Scout's items are unchecked.** Its list is longer than a human panel's, and part of its hit rate is volume; how many of its items are wrong or already answered by the paper was not measured.
- **Compressed ratings.** On the 86 papers, 78% receive final Overall 3 (ICLR 4); accepted papers are rated 0.8 points below the humans. Better ranking at the top and bottom of the pool is the main open problem.
- **Human labels are noisy.** On the 86 papers, one human agrees with another at only Spearman 0.14. Mean human ratings are a weak target, and acceptance is a coarse one.

Reproducibility

The skill, the scoring script, the evaluation harness, per-paper scores and paper IDs are in the repository. Manuscripts and review text are not committed. Key commands:

```
python3 -m unittest discover -s tests -v
python3 evals/iclr_calibration.py features RESULTS_DIR --out features.jsonl
python3 evals/iclr_calibration.py pilot sample.jsonl features.jsonl
python3 evals/iclr_calibration.py evaluate sample.jsonl features.jsonl \
        --splits splits.json
```

Paid runs use the workflow `.github/workflows/iclr-calibration.yml`, configured by `evals/iclr_run.json`.

References

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A Per-paper pilot scores

		Human	Haiku		Sonnet high		
Paper	Decision	mean	panel	pool	panel	pool	Recognized
AIKURJjxXT	Withdrawn	2.50	3.2	40	2.8	40	
KtocialdxS	Reject	2.67	3.4	48	3.0	45	
8xcmfwomnI	Reject	3.00	3.0	40	3.0	42	
g4HpTtgsn2	Withdrawn	3.00	3.2	46	2.6	38	
E5mLO7Fbq8	Reject	3.33	3.0	38	2.6	42	
Z6z4AnXGhw	Reject	3.50	3.4	50	3.2	50	
gUw8n109ki	Withdrawn	3.50	3.2	50	2.6	40	
0xObmDmsKY	Reject	4.00	3.2	46	3.0	45	
BxeJLOrKDF	Withdrawn	4.00	3.2	45	3.2	45	
Jw7khYzYzl	Reject	4.00	3.6	45	4.0	70	✓
9jcB40wjk3	Accept	4.40	2.8	40	2.8	38	
be4ey66fhk	Reject	4.40	3.2	45	3.4	52	
9R2iUHhVfr	Reject	4.50	3.4	50	3.6	65	✓
PJdMrK79Mo	Accept	4.50	3.6	48	3.4	50	
qfW4IrUuig	Reject	4.50	3.0	40	2.8	40	
LvkbeOCgmZ	Reject	5.00	3.4	40	3.2	45	
tlYSbw5GXY	Accept	5.20	3.2	38	3.4	55	
euMVC1DO4k	Accept	6.00	3.6	50	3.8	68	✓
kwro5432AI	Desk rej.	6.00	3.2	45	3.0	42	
rubeJmT1XM	Accept	6.00	3.2	40	3.0	40	

Human mean is on the ICLR 2026 scale (0–10). "Panel" is the reviewers' mean Overall (1–6) and "pool" is the EIC pool position (0–100).